

The Biological Compiler: raw notes

RFdiffusion → ProteinMPNN → ESMFold, on one GPU, targeting a
PFAS-adjacent enzyme

everything in here is either measured or explicitly marked as a guess
nothing is invented to sound better

The Problem

PFAS contamination is expensive, and it's getting more expensive. There isn't a biological solution deployed anywhere near the scale of the problem. The EPA's 2024 drinking water rule set hard limits, 4.0 parts per trillion for PFOA and PFOS, with monitoring required by 2027 and compliance by 2029, and the cost numbers that already exist are real: \$3 billion a year projected in wastewater treatment across affected industries, \$62 billion-plus in healthcare costs tied to PFAS-linked disease, settlements already landing in the millions. Every company actually working on this that could be verified is doing physical or chemical treatment: filtration, adsorption, supercritical water oxidation, UV-based destruction. None of them are trying to design an enzyme that just breaks the bond. That's the opening this project is aimed at, whether or not it turns out to be a real one.

What Exists – Competitive Landscape

Four companies, each verified directly against their own site. Not fifteen. Four real ones beats a padded list.

Company	Tech	Approach
374Water	AirSCWO (supercritical water oxidation)	Physical/chemical destruction. Up to 99.9999% reduction claimed.
Cyclopure	DEXSORB® adsorbent	Filtration, not destruction.
Claros Technologies	ClarosTechUV™	Undisclosed mechanism, likely UV-based destruction.
Allonnia	SAFF® foam fractionation	Physical concentration. PFAS is a secondary product line; their core business is mining biotech.

RAW NOTE — this is not a comprehensive competitive landscape. it's what could be verified with the search budget available this session. treat it as a floor, not a ceiling.

The Gap

Protein design tools: RFdiffusion, ProteinMPNN, ESMFold, etc. (exist, open-source, this project uses them directly)

PFAS remediation products: 374Water, Cyclopure, Claros, Allonnia (exist, physical/chemical, deployed)

MISSING: an actual pipeline connecting AI-designed enzymes to PFAS-specific targets to wet-lab validation to a real product

The tools exist. The remediation companies exist. Nobody's built the middle layer end to end, from generation through an actual assay. That's not proof it's a good idea. It's just an observation about what's missing.

Our Pipeline

Real output, not a mockup. This is what a single RFdiffusion + ProteinMPNN + ESMFold cycle actually printed:

```
[10:21:01] Loading checkpoint.
Successful diffuser __init__
[10:21:06] Making design smoke_test_outputs/smoke_0
Timestep 50 ... Timestep 2 ...
[10:21:42] Finished design in 0.43 minutes

20 sequences of length 184 generated in 2.87 seconds

seq3: pLDDT=67.4 RMSD=3.87A
seq7: pLDDT=71.4 RMSD=4.43A
=== RUNNING TOTALS: 900 sequences evaluated, 298 candidates passed ===
```

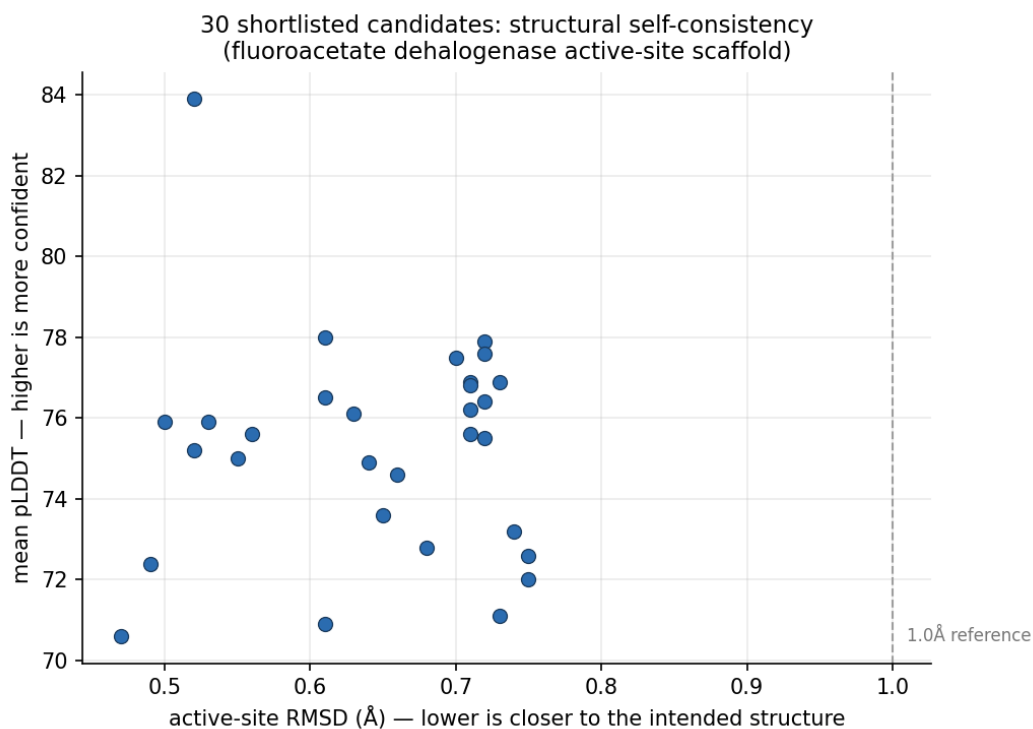
The core scoring function, the thing that decides whether a design counts as a "candidate," is a Kabsch alignment plus an RMSD check restricted to the catalytic residues:

```
def active_site_rmsd(reference_pdb, predicted_pdb, motif_positions):
    ref, pred = parse_ca(reference_pdb), parse_ca(predicted_pdb)
    pred_aligned = kabsch(pred_coords, ref_coords)
    idx = [ref_resnums.index(p) for p in motif_positions]
    return rmsd(pred_aligned[idx], ref_coords[idx])
```

Full code lives in [pipeline/](#). Not a demo; this exact logic scored all 3,600 real sequences behind the shortlist below.

The Shortlist

30 candidates, curated from 1,253 that passed $\text{pLDDT} \geq 65$ / $\text{RMSD} \leq 2.0 \text{ \AA}$ out of 3,600 total, capped at 3 per backbone so a few lucky designs don't dominate. Every one below is sub-angstrom on active-site RMSD.



#	ID	pLDDT	RMSD (Å)	len
1	batch2_design_0073_seq5	70.6	0.47	186
2	batch1_design_0098_seq3	72.4	0.49	181
3	batch2_design_0152_seq6	75.9	0.50	232
4	batch2_design_0049_seq5	75.2	0.52	240
5	batch2_design_0152_seq1	83.9	0.52	232
...	25 more in data/shortlist/manifest.csv; full sequences included there, omitted here for space			
30	batch1_design_0042_seq3	72.6	0.75	155

What's Real vs. What's Guess

Claim	Status
Pipeline runs end-to-end on an RTX 5090	REAL. Run twice, 400 backbones, 3,600 sequences total
30 candidates are structurally self-consistent	REAL. Measured RMSD/pLDDT, not estimated
FAcD catalytic residues (Asp104/Asp128/His271)	REAL. Verified against the downloaded PDB file directly
Any candidate actually degrades PFAS	GUESS. Untested, unsynthesized, zero wet-lab data
FAcD is a good PFAS proxy target	An informed guess. Literature-endorsed family, wrong substrate
\$132B liability figure (earlier draft)	DROPPED. Could not verify anywhere
15+ mapped competitors (earlier draft)	DROPPED. Only 4 could be verified
rdhA is the "real" PFAS enzyme	REAL biology, but structurally unusable. No public sequence anywhere checked

Next Steps

Specific and bounded, not "change the world":

Action	Owner	Bound
Send outreach email to Jaffé and Wackett (closest-fit PIs)	founder	done, 2026-08-15
If a lab responds: ship 5–10 sequences for cell-free expression + fluoride assay	founder + lab	whenever a lab is ready, not before
Recheck UniProt/NCBI/AlphaFold DB for a public rdhA structure	founder	monthly
Do NOT start building FastAPI/React/Docker platform	founder	until something is validated

RAW NOTE — the software is not the bottleneck anymore. the bottleneck is getting one sequence into an actual test tube. everything else is optional until that happens.